CSCI1409 Deep Learning | Aidyn Abirov | Week 6 Tasks 1–3 – Estimated time: about 1.5–2 hours – **Deadline:** before the next lecture – **Submit:** notebook via MS Teams Assignments

This homework is about using **pretrained models**. You will run ResNet18, a CNN trained on 1.2 million ImageNet photos, look inside it, use it as a feature extractor for image search, and then adapt it to a new task — telling ants from bees — with **transfer learning**. Everything happens in the provided notebook `CSCI1409_Week06_Homework.ipynb`: fill in every `# TODO`, and use the `assert` lines to check your work.

Rules

- Work individually.
- **Tasks 1–2: no AI** (search engines, lecture notes, textbooks, and the PyTorch documentation are fine – list references). Do not use AI-generated search summaries.
- **Task 3: AI allowed** only to explain concepts or help debug Task 3 code. State which tool you used and for what. You must understand and be able to defend every submitted line.
- Do not change the `assert` lines. `cosine_similarity` in Task 2 must not call `F.cosine_similarity`.
- Use PyTorch, torchvision, NumPy, and Matplotlib only.
- Submit `Surname_Firstname_Homework06.ipynb` via the Assignments tab on Teams; restart the kernel and run all cells first. No submissions once the next lecture starts.

Setup. Same as Homework 5 (`pip install torch torchvision matplotlib`, or Google Colab). The first run downloads the ants/bees images and the ResNet18 weights (about 45 MB each), so it needs internet. A GPU is not needed: all training together takes a few minutes on a laptop CPU.

Dataset. About 240 training and 150 validation photos of ants and bees, in folders `hymenoptera_data/train/ants`, `.../bees`, and the same for `val`. This is deliberately small: the question of the week is what you can do with little data.

Task 1 – Use a pretrained model as it is

About 25 minutes. No AI.

1. **Normalisation by hand:** normalise one RGB pixel with the ImageNet mean and std.
2. **Predictions:** classify four photos with ResNet18 and print the top-5 ImageNet classes (softmax + `torch.topk`).
3. **Without normalisation:** run the same photos without `Normalize` and compare.
4. **Look inside:** find the last layer, count its parameters by hand, and count the whole model (11.7 million vs. 9 thousand in Homework 5).
5. **Filters:** compare ResNet18's 64 colour filters with the 8 you learned on MNIST.
6. Answer Questions 1.

Task 2 – Pretrained features and CNN applications

About 25 minutes. No AI.

1. **Cosine similarity by hand** for three pairs of small vectors; then code it from scratch.
2. **Feature extractor:** replace ResNet18's last layer with `nn.Identity()` so every image becomes 512 numbers.
3. **Similarity:** are ants closer to ants than to bees?
4. **Image search:** show the 5 photos most similar to a query photo — with no training at all.
5. **Applications table:** for six real applications, decide whether the task is classification, detection, or segmentation, and what the model outputs.

6. Answer Questions 2.

Task 3 – Transfer learning vs. training from scratch

About 45 minutes. AI assistance allowed, with individual defense.

Train three models on the same images and compare them:

run	what is trained	learning rate	epochs
from scratch	a small CNN like Homework 5 (count its parameters by hand)	0.01	10
feature extraction	ResNet18 frozen; only a new `Linear(512, 2)`	0.01	5
fine-tuning	as above, plus ResNet18's last block (`layer4`)	0.005	5

You write the freezing (`requires_grad = False`), the new last layer, and the unfreezing of `layer4`. The training loop is given. The notebook then prints a comparison table (trainable parameters, time, accuracy), plots accuracy per epoch, and shows the misclassified photos. Answer Questions 3. *Optional:* add data augmentation to the from-scratch model.

In the defense, be ready to explain any line of your code, for example: why the new last layer is trainable without setting anything, what `nn.Identity()` does, and why fine-tuning uses a smaller learning rate.

Submission checklist

- Every `# TODO` filled, every `assert` passes, and the notebook runs from a fresh kernel (Restart & Run All)
- All **Your answer** cells written (Questions 1, 2, 3 and the applications table)
- Hand calculations for 1a, 1d, 2a, and 3a done on paper first
- References listed; Task 3 AI use recorded (or "no AI used"); no AI used for Tasks 1–2
- Submitted via Teams Assignments before the next lecture begins

Reminder: a working notebook does not replace the individual defense during practice.